

Original Article

Development and temporal validation of a deep learning model for automatic fetal biometry from ultrasound videos

M. Goetz-Fu^{a,*}, M. Haller^{b,1}, T. Collins^{b,c}, N. Begusic^a, F. Jochum^a, Y. Keeza^c, J. Uwineza^c, J. Marescaux^b, A.S. Weingertner^a, N. Sananès^{a,2}, A. Hostettler^{b,c,2}

^a Department of Obstetrics and Gynecology, Strasbourg University Hospital, Strasbourg, France

^b Research Institute against Digestive Cancer (IRCAD), Strasbourg, France

^c Research Institute against Digestive Cancer (IRCAD), Kigali, Rwanda

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ABSTRACT

Objectives: The objective was to develop an artificial intelligence (AI)-based system, using deep neural network (DNN) technology, to automatically detect standard fetal planes during video capture, measure fetal biometry parameters and estimate fetal weight.

Methods: A standard plane recognition DNN was trained to classify ultrasound images into four categories: head circumference (HC), abdominal circumference (AC), femur length (FL) standard planes, or 'other'. The recognized standard plane images were subsequently processed by three fetal biometry DNNs, automatically measuring HC, AC and FL. Fetal weight was then estimated with the Hadlock 3 formula. The training dataset consisted of 16,626 images. A prospective temporal validation was then conducted using an independent set of 281 ultrasound videos of healthy fetuses. Fetal weight and biometry measurements were compared against an expert sonographer. Two less experienced sonographers were used as controls.

Results: The AI system obtained a significantly lower absolute relative measurement error in fetal weight estimation than the controls (AI vs. medium-level: $p = 0.032$, AI vs. beginner: $p < 1e-8$), so in AC measurements (AI vs. medium-level: $p = 1.72e-04$, AI vs. beginner: $p < 1e-06$). Average absolute relative measurement errors of AI versus expert were: 0.96 % (S.D. 0.79 %) for HC, 1.56 % (S.D. 1.39 %) for AC, 1.77 % (S.D. 1.46 %) for FL and 3.10 % (S.D. 2.74 %) for fetal weight estimation.

Conclusion: The AI system produced similar biometry measurements and fetal weight estimation to those of the expert sonographer. It is a promising tool to enhance non-expert sonographers' performance and reproducibility in fetal biometry measurements, and to reduce inter-operator variability.

Introduction

Fetal biometry measurements are performed routinely by sonographers, to reveal growth anomalies that often point to pregnancy-related pathologies, which must be correctly handled and may elicit a birth decision. The Hadlock 3 [1] formula is the most widely used formula to calculate fetal weight using three biometry parameters: head circumference (HC), abdominal circumference (AC), and femur length (FL).

International guidelines have been established to promote standardization by consortiums such as the International Fetal and Newborn Growth Consortium for the 21st Century Initiative (INTERGROWTH)

[2] and the International Society of Ultrasound in Obstetrics and Gynecology (ISUOG) [3]. However, intra-operator variability for HC, AC, and FL measurements can be as high as 3.3 %, 5.4 %, and 6.8 % respectively, while inter-operator variability reaches 5.1 %, 8.9 %, and 12 % respectively [4]. The learning curve across residency illustrates that substantial training is needed for humans to achieve sufficient accuracy, and accuracy increases with greater experience.

Measurement errors can induce misclassification of small-for-gestational age or large-for-gestational age fetuses [5,6]. Crucial clinical decisions are made based on fetal weight estimations, including induction of labor and cesarean section for suspected macrosomia. A

* Corresponding author at: Department of Obstetrics and Gynecology, Femme-Mère-Enfant University Hospital, Hospices Civils de Lyon, France.

E-mail address: morgane.goetz9@gmail.com (M. Goetz-Fu).

¹ Co-first authors.

² Co-last authors.

misdiagnosis of fetal growth restriction is highly concerning as it may induce prematurity, which is the leading cause of neonatal morbidity and mortality around the world [7]. Diagnosis errors in fetal weight estimation may lead to unnecessary interventions and induce stress for the parents-to-be.

Artificial intelligence (AI) has the potential to improve fetal biometry by reducing operator dependence and to make routine fetal biometry more accessible in remote regions of the world where there is a lack of trained sonographers.

Deep learning AI is currently the best technology for automatic image analysis and image recognition [8]. Our objective was to develop and test, in a prospective temporal validation study, the performance of a deep learning AI-based system that detects standard fetal planes in ultrasound video loops and measures fetal biometry parameters.

We followed CONSORT-AI statements to ensure the quality and reproducibility of our work [9].

Methods

Data sources

Training data

The AI training phase relied on two independently collected sources of training data. Training Data Source 1: Our team retrieved still images from a publicly available dataset created in two hospitals in Barcelona, Spain (Hospital Clinic and Hospital Sant Joan de Deu), by multiple operators during routine clinical practice in large fetal-maternal departments performing thousands of deliveries each year [10]. As described in Burgos-Artizzu et al., patients with gestational ages between 18 and 40 weeks were included, and multiple pregnancy, congenital malformations or aneuploidies were excluded. We included all images from Training Data Source 1 labeled as AC, FL and HC standard planes, necessary to use the Hadlock fetal weight estimation formula.

Training Data Source 2: This was acquired prospectively by one sonographer during routine screening sonographies in two French echography centers based in Strasbourg, France (Centre Médico-Chirurgical et Obstétrical and Centre Hospitalo-Universitaire de Haute-pierre) between July 2020 and March 2021. Ultrasound scans of fetuses aged from 15 to 35 weeks of pregnancy were included. Congenital malformations and aneuploidies were excluded. All patients were informed that their data would be registered, anonymized, and used for a scientific study. Under French regulations, this study is exempt from Institutional Review Board review because it is an observational study using anonymized data from medical records.

Training datasets, used to train the system's deep neural networks, were constructed from the training data sources as follows:

- The Standard Plane Recognition Training Dataset

An experienced sonographer (~4 years) validated the images in both data sources according to the reference definitions of standard planes [3]. Several images were found to be incorrect: for example, some images labeled as FL standard planes were images of fetal radiuses. All such images were presented to an expert gynecologist (~20 years), and any incorrectly labeled images were removed.

The Standard Plane Recognition Training Dataset thus comprised 1220 AC images, (329 from Source 1, and 891 from Source 2), 1601 images of HC images (1005 from Source 1 and 596 from Source 2), and 1073 FL images (611 from Source 1, and 462 from Source 2). Finally, 9732 images were considered as “non-standard planes” (7339 from Source 1 and 2393 from Source 2).

- The AC Measurement Dataset

This dataset consisted of 650 manually annotated AC images from

the Standard Plane Recognition Training Dataset (300 images from Source 1 and 350 from Source 2). The sonographers placed five landmarks around the outer layer of the abdominal wall at the anatomical level comprising the portal sinus, one adrenal gland, and the stomach bubble if visible, using the CVAT computer vision annotation platform [11] (Fig. 1).

- The HC Measurement Dataset

This dataset consisted of 695 manually annotated HC images from the Standard Plane Recognition Training Dataset (580 from Source 1 and 115 from Source 2). The sonographers annotated four landmarks using the CVAT annotation tool, placed at the outer layer of the cranial bone using the transthalamic plane.

- The FL Measurement Dataset

This dataset consisted of 1060 manually annotated FL images from the Standard Plane Recognition Training Dataset, including 580 images from Source 1 and 460 images from Source 2. The sonographers annotated two anatomical landmarks using the CVAT annotation tool, corresponding to the ends of femoral diaphysis.

We summarize the composition of the training datasets in Fig. 1.

Evaluation dataset

The Evaluation Dataset consisted of 281 anonymous video sequences collected at CMCO of 80 patients, between August 2021 and March 2022, using General Electrics Voluson E10 devices, of fetuses aged from 15 to 35 weeks and exempt from malformation. It comprised 80 AC and 80 FL videos (one per patient), and 79 HC videos from 79 patients (one per patient). For one patient, an HC video was not recorded, and consequently, HC and weight accuracy could not be assessed for that patient. Each video was recorded continuously to avoid selection bias, with an average duration of 6.48 s (S.D. 3.32 s), with extreme values from 1.35 to 15.12 s.

The most experienced sonographer was defined as the gold standard (or ‘ground-truth’). Quality assessment of standard planes were defined by Salomon et al. [12].

Dataset pre-processing

The datasets were pre-processed as follows:

1. File conversion and anonymization: Images from Training Data Source 1 were anonymized by the authors of the dataset. Images from Training Data Source 2 were anonymized to remove all sensitive and patient-identifiable meta-data as follows. First, they were converted from DICOM to the Portable Network Graphics (PNG) image format, which provided lossless compression and removed all DICOM fields. Next, all sensitive text appearing in the images was digitally retracted. Because the data came from a limited number of devices, and whose sensitive text appeared in consistent locations, the digital retraction could be automated, and all redactions were then manually verified. The evaluation videos were converted from DICOM to the H. 265 format in lossless compression, which also removed all DICOM fields. All sensitive text from every video frame was then digitally retracted similarly to the images from Source 2.
2. Grayscale conversion. The images and videos were then, if required, converted from color to grayscale using the OpenCV Computer Vision software library (version 4.5.5.64). This conversion was performed because B-mode ultrasound images are monochrome, so colour channels do not convey useful information.
3. Spatial resizing. All images in the training datasets were spatially downsized to fixed resolutions: Images in the Standard Plane Recognition Dataset were downsized to 400×400 pixels, and images in the AC, FL, and HC Datasets were downsized to 448×448 pixels.

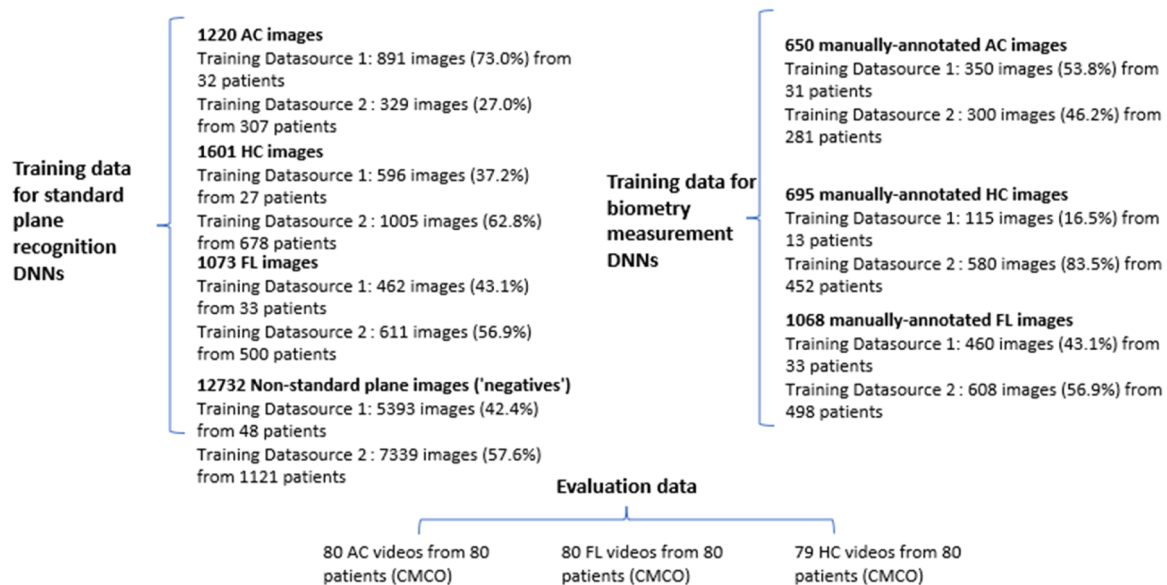


Fig. 1. Description of the datasets. Two different sources of images were used as described above: Training Data Source 1 refers to the French sonography departments (including CMCO and Haute-pierre teaching hospital) and Training Data Source 2 refers to the online database from Barcelona, Spain. We separate the training dataset and the evaluating dataset. The training dataset was cleared from non-standard planes by expert readers resulting in the exclusion of the “non-standard plane images”. AC= abdominal circumference, HC= head circumference, FL= femur length.

Model development

Each DNN was trained independently on the training sets as described above. For method reproducibility, the complete training configurations of each DNN described in this section, which include data augmentation, training, hyperparameters, software libraries, library versions, and computational hardware, are supplied in Appendix A. An overview of the workflow showing how the DNNs work together is provided in Fig. 2.

Standard plane recognition DNN

Using the Standard Plane Recognition Training Dataset, a DNN was trained to automatically classify an ultrasound image into four categories: AC standard plane, FL standard plane, HC standard plane, or ‘other’ (meaning any ultrasound image that should not be considered as a standard plane). To this end, the EfficientNetB2 architecture was used, based on its excellent performance in solving image recognition tasks. This was adapted to four-class classification using one classification head with four output neurons with Softmax activation.

The EfficientNetB2 was trained using the Python 3.8.12 software language and the Catalyst DNN training framework (22). The model was trained using Adam optimization, using the Cross-entropy loss function with class-weighted balancing. Early stopping was applied to reduce the risk of overfitting, set to 200 epochs. To overcome the limitations associated with training DNNs on smaller datasets, the techniques of model fine-tuning and data augmentation were both employed. Fine-tuning refers to training a model using pre-defined starting weights, established from a larger image recognition dataset (ImageNet). Data augmentation refers to virtually enlarging the training dataset by adding random perturbations to each training image (in our case, we applied variations in image contrast, rotation angles, and structure depths). The reader is referred to Appendix A for all training hyperparameters.

AC Biometry Estimation DNN and AC Measurement

The 2nd DNN was independently trained to automatically delineate (‘segment’) the abdomen in an AC standard plane image, using the AC Measurement Training Dataset. This delineation, also known as an image segmentation, was represented as a binary mask of the same size as the standard plane image (448×448 pixels), where mask values of 0 and 1 represented pixels that were automatically recognized as being

outside, and inside the fetal abdomen, respectively. The DNN was implemented using a state-of-the-art neural network architecture, called UNet++, selected based on its strong performance in related medical image segmentation tasks.

The UNet++ was adapted to binary segmentation using a two-channel output layer (one for the background class and one for the foreground class), with Softmax activation. The UNet++ model was trained by fine-tuning its initial weights on the training set’s reference AC segmentation masks. Adam optimization was used to minimize the Dice loss, with data augmentation, and early stopping after 100 epochs.

After training, the DNN was used for AC measurement as follows. Firstly, an AC standard plane image was inputted to the DNN (set to inference mode), and the activations of the output layer neurons were thresholded (above 50 % confidence) to produce a binary 2D mask (or ‘segmentation mask’). Next, the mask’s perimeter was extracted using the OpenCV software library’s findContours method. Next, an ellipse was automatically fitted to the perimeter using the algorithm of Fitzgibbon et al. (23). Finally, the AC measurement was computed by calculating the circumference of the fitted ellipse.

HC biometry estimation DNN and HC Measurement

The 3rd DNN was independently trained to automatically delineate the cranium in an HC standard plane image, using the HC Measurement Training Dataset. The UNet++ architecture was used, adapted to binary segmentation using a two-channel output layer with Softmax activation, similarly to the AC Biometry Estimation DNN. The UNet++ model was trained by fine-tuning its initial weights on the 695 training set’s reference HC segmentation masks. Adam optimization was used to minimize the Dice loss, with data augmentation, and early stopping after 100 epochs.

After training, the DNN was used for HC measurement using the same processing steps as AC measurement: Firstly, an HC standard plane image was inputted to the DNN, and from the resulting segmentation mask, the perimeter was extracted. An ellipse was fitted to the mask, and the HC measurement was computed as the circumference of the fitted ellipse.

FL biometry estimation DNN

The 4th DNN was trained to automatically localize the femur end-

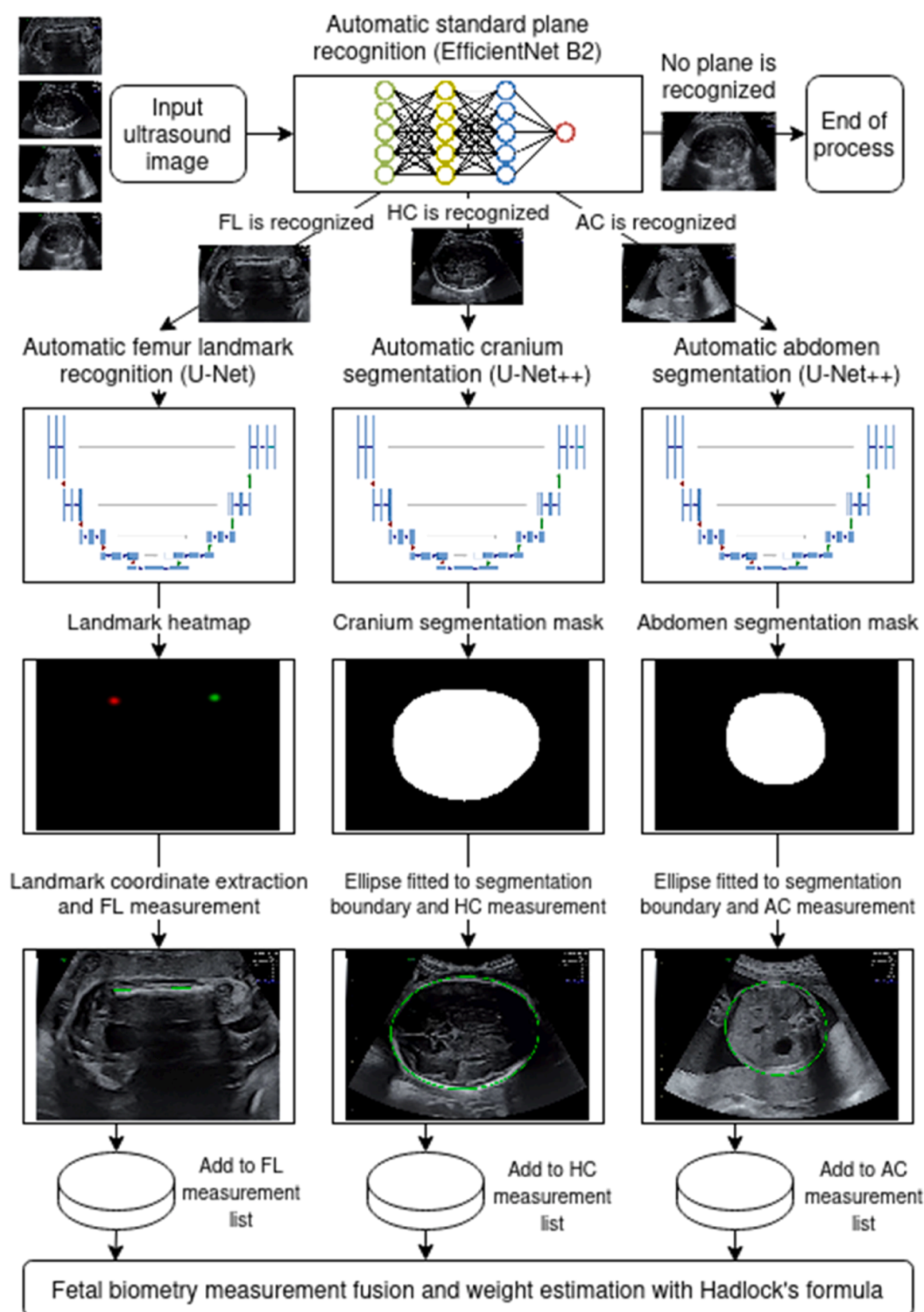


Fig. 2. Workflow of the presented fetal biometry AI system. Each frame (image) from the ultrasound video is processed by the first DNN, which automatically classifies the image into four categories (FL standard plane, HC standard plane, AC standard plane, or non-standard plane). Classified standard plane images are then processed by subsequent DNNs, which extract the visual information required to make biometry measurements. The measurements are then added to a list, and once FL, HC, and AC measurements are obtained, weight is estimated using measurement fusion (median value) and Hadlock's formula. DNN: Deep Neural Network, US: ultrasound.

point landmarks in an FL standard plane image. Training was achieved using supervised learning, with a so-called heatmap representation. In this context, a heatmap is an image with the same resolution as the ultrasound image, storing intensity values at each pixel, where a brighter value is associated with a pixel closer to the landmark (Fig. 3). This was implemented using a binary heatmap, represented as a 2D circular mask centered on each of the landmark's 2D positions. The DNN was then trained to predict the masks. The UNet architecture was used, trained using the Dice loss. Training was performed with Adam optimization, with data augmentation and early stopping after 100 epochs.

After training, the DNN was used for FL measurement as follows. Firstly, an FL standard plane image was inputted to the DNN (set to inference mode), and the predicted heatmap's highest-scoring 2D locations were used as predictions of the femur end-points. Next, the FL measurement was computed as the Euclidean distance between the femur end-points.

Fetal weight estimation from videos

This section describes how the DNNs and biometry estimation methods described in the previous sections were configured to automatically estimate fetal weight from a patient's ultrasound video data. The processing steps are illustrated in Fig. 2. The video data may consist of a single video, or a set of videos featuring different standard planes. First, every video frame is sequentially inputted into the standard plane recognition DNN. Next, all images recognized as AC, FL, or HC standard planes are inputted into the AC, FL, and HC biometry estimation DNNs. Next, the DNN outputs are converted to biometry measurements as described in the above sections, producing a set of biometry measurements. Note that more than one biometry estimate can be made, occurring if more than one standard plane image is recognized throughout the video(s). Next, the median biometry measurement is computed, and finally, fetal weight is estimated from the median measurements using the Hadlock 3 formula.

Evaluation strategy

Temporal validation methodology

Performance was evaluated using the Evaluation Dataset using a prospective temporal validation methodology [13]. Temporal validation

assesses performance on data collected from the same center as training data, at a different time than the training data. The validation data comprised new patients (i.e. the same patients do not appear in the training dataset). The main strength of temporal validation is to allow performance generalization to be assessed on different patients, and in a future time period. Temporal validation is often regarded as a validation approach used in AI model development, which lies midway between internal and external validation [14].

Gold standard and controls

The biometry parameters were assessed in each video independently by 3 gynecologists trained in fetal sonography ('annotators'), ranging in fetal biometry experience from 1 year of experience (beginner, Control 1), 4 years of experience (medium level, Control 2), and 20 years (expert, Gold Standard). The annotators used the CVAT annotation software, installed on a standard desktop PC, to process ('annotate') each video as follows. First, the entire video was inspected frame-by-frame, and then the highest-quality standard plane image in each video was selected. Next, the annotator marked the associated biometry information in the selected image using CVAT's annotation tool (as illustrated in Fig. 3). Finally, all annotation data was exported from CVAT using CVAT's XML file convention. To ensure an independent evaluation, the annotators did not communicate with each other during the annotation process.

Metrics and outcomes

The primary metric was the fetal weight absolute relative error (ARE) (%) in fetal weight (one sample per patient). The primary outcome was the average difference in fetal weight ARE, comparing the AI system against controls. The secondary metrics and outcomes were as follows:

1. Fetal biometry measurement error. Metrics: AC, FL, and HC ARE (%). Outcomes: average difference in AREs comparing the AI system against controls.
2. Inter-operator agreement. Metrics: Intraclass correlation coefficient (ICC – type 1–1). Outcomes: ICC values.

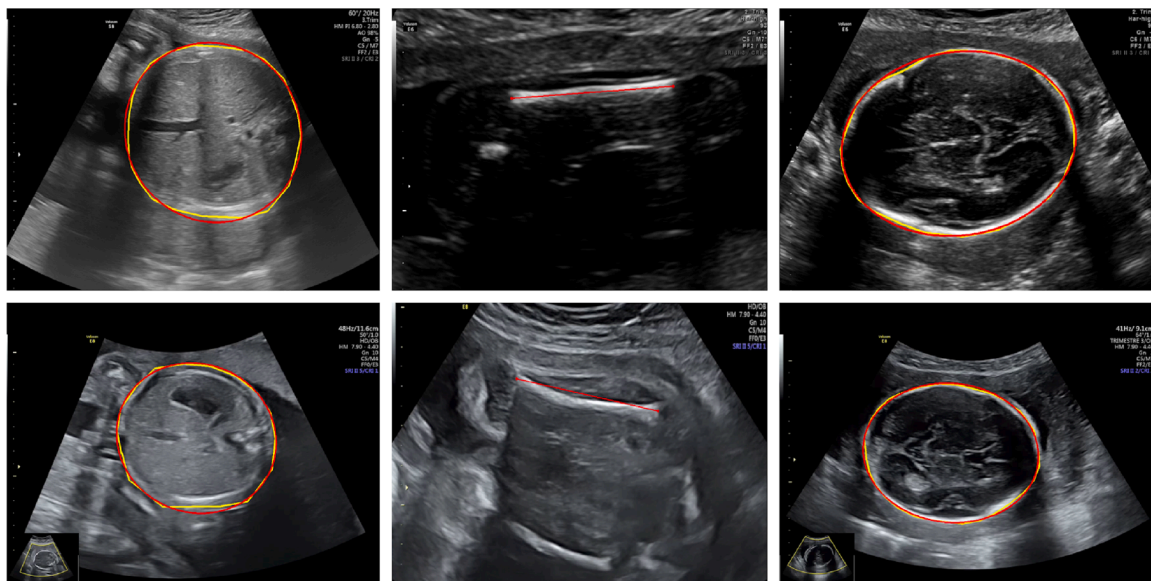


Fig. 3. Illustration of annotated abdominal 2D ultrasound images used to train the DNNs (axial plane for AC and HC, frontal plane for FL). The top row shows AC (left), FL (middle) and HC (right) annotations of a randomly selected patient from the public dataset. For the AC and HC annotations, the yellow border shows the border of the segmentation annotation, and the red border shows the best-fitted ellipse to the segmentation data. The bottom row shows AC (left), FL (middle) and HC (right) annotations of a randomly selected patient from the CMCO ultrasound department.

AI configuration

The AI system was configured to mimic the controls as follows. For each patient in the Validation dataset, the patient's video data was processed as described in Methods, "Fetal Weight Estimation from Videos", to estimate the patient's weight, AC, FL, and HC measurements.

To classify images with the Standard Plane Recognition DNN, detection confidence thresholds had to be set. As with all AI model development, setting an appropriate confidence threshold was important, involving a trade-off between accuracy and sensitivity. In this case, three separate confidence thresholds were defined—one for each standard plane. Increasing the confidence threshold generally diminished measurement discrepancies but reduced the sensitivity of plane detection. To determine confidence thresholds, we conducted an empirical analysis by incrementally adjusting the confidence values from 0.0 to 1.0 and assessing their impact on sensitivity and measurement error. Specifically, we identified threshold values corresponding to the 'elbow point' on sensitivity-error graphs, where further increases in sensitivity led to a sharp rise in measurement error. These elbow points, which indicated an appropriate balance between sensitivity and accuracy, were used to set the confidence thresholds for the AC and FL plane detection. However, no clear elbow point was observed for the HC plane.

In this case, we selected the lowest threshold that achieved 100 % sensitivity. The sensitivity-error trade-off graphs, illustrating these selections, are provided in Appendix B. To illustrate the confidence interval of measurements, examples of low-fitted and best fitted measurements are provided in Fig. 4.

Statistical analysis

Statistics were computed with the statistical toolbox of Matlab 2023a.

Primary outcome and Secondary outcome 1: Paired two-tailed *t*-tests were used to compare the mean of differences in ARE. We set an alpha-risk at 5 % ($p = 0.05$) to define the significance threshold.

Results

Of the 80 patients with videos containing AC, and FL standard planes, the AI system automatically recognized at least one AC and FL standard plane image for 77 (96.3 %) and 79 (98.7 %) patients respectively. Of the 79 patients with videos containing HC standard planes, the AI system automatically recognized at least one HC standard plane

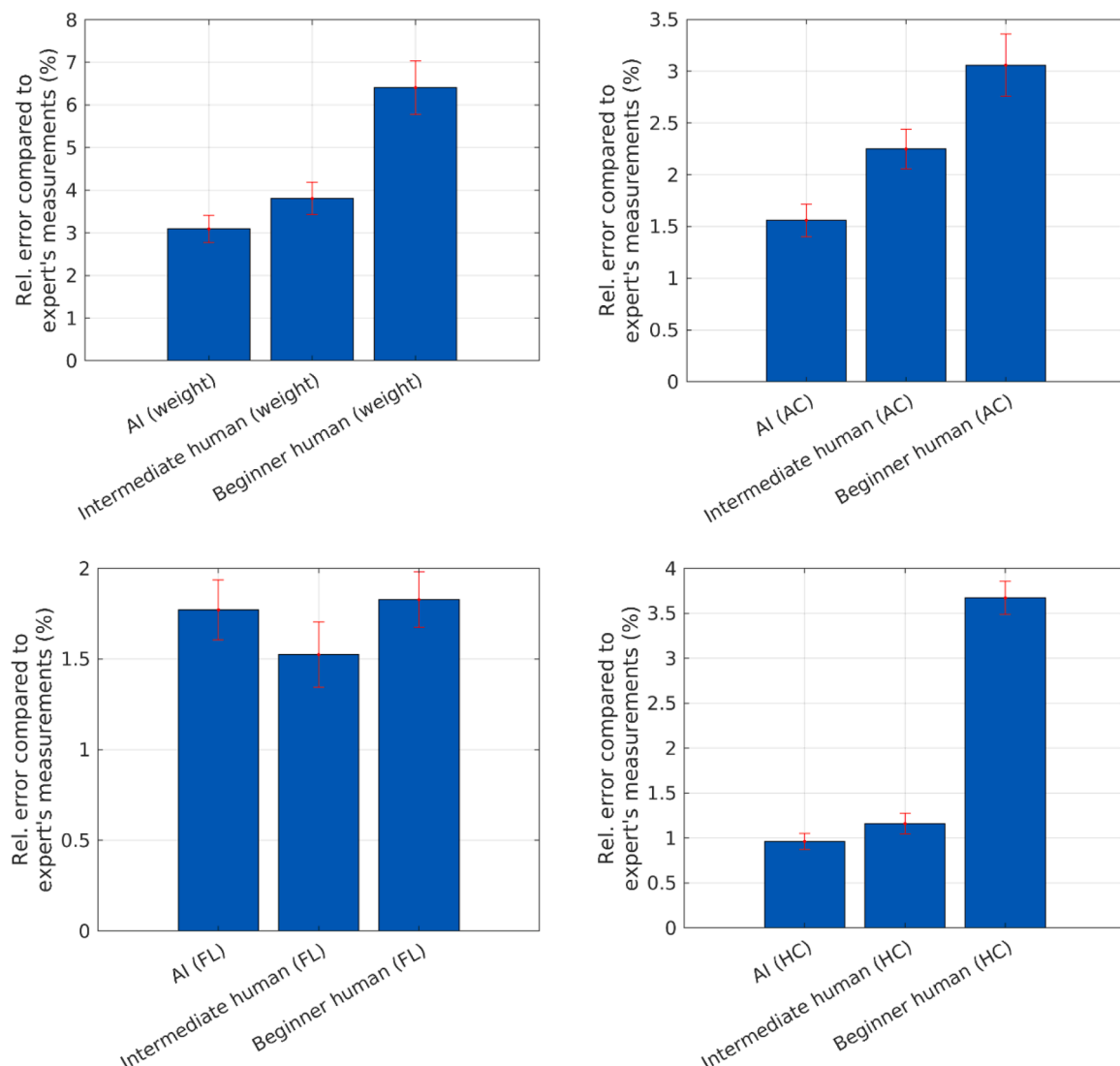


Fig. 4. Accuracy of fetal parameters automatically estimated from ultrasound videos using the AI system, compared to three obstetrician-gynecologist sonographers. Top left: bar charts showing the mean absolute relative error in fetal weight estimates compared to the expert gynecologist. The three bars show the performance of the AI system, the medium-level experienced sonographer, and the beginner sonographer. Top right, bottom left, and bottom right: equivalent bar charts showing the mean absolute relative error in biometry estimation for abdomen circumference (AC), femur length (FL), and head circumference (HC).

image for 79 (100 %) of the patients. Of the 79 patients with videos containing AC, FL, and HC standard planes, the AI system automatically recognized all standard plane images for 75 (95 %) of patients.

The results of the AI's biometry measurement performances are summarized in Fig. 5. This shows bar charts of the mean absolute relative error (MARE) (%) of fetal weight, AC, FL, and HC measurements, compared to the measurements of the expert. MARE performance of the AI system for fetal weight was 3.10 % (S.D. 2.74 %), compared to 3.81 % (S.D. 3.28 %), and 6.41 % (S.D. 5.42 %) for the medium-level and beginner. The differences were statistically significant (AI vs. Medium-level: $p = 0.032$, AI vs. beginner: $p < 1e-8$). Inter-rater agreement was also measured using the intraclass correlation coefficient (ICC), showing scores of 0.994 (AI vs. expert), 0.991 (medium-level vs. expert), and 0.974 (beginner vs. expert).

For AC measurement performance (MARE): the AI system achieved 1.56 % (S.D. 1.39 %), compared to 2.50 % (S.D. 1.69 %), and 3.06 % (S.D. 2.63 %) for the medium-level and beginner. The differences were statistically significant (AI vs. Medium-level: $p = 1.72e-04$, AI vs. beginner: $p < 1e-06$).

For FL measurement performance (MARE): the AI system achieved 1.77 % (S.D. 1.46 %), compared to 1.52 % (S.D. 1.61 %), and 1.83 % (S.D. 1.36 %) for the medium-level and beginner. The differences were not statistically significant (AI vs. Medium-level: $p = 0.20$, AI vs. beginner: $p = 0.78$).

For HC measurement performance, MARE the AI system was 0.96 % (S.D. 0.79 %), compared to 1.16 % (S.D. 1.03 %), and 3.67 % (S.D. 1.64 %) for the medium-level and beginner. The difference between AI vs. beginner was statistically significant: $p < 1e-06$, but AI vs. Medium-level was not.

The average computational time to process each image using all 4 DNNs was 48 frames per second. The time required to compute fetal weight from the DNN outputs was in the range of a few milliseconds. These computational times were obtained on a computer equipped with an RTX 3090 24GO VRAM GPU, an AMD Ryzen 9 5950 × 16-Core CPU, 64Go of DDR4 2667 MHz RAM, and a KXG70PNV2T04 NVMe KIOXIA 2048GB SSD disk drive, running on the Ubuntu 22.04 operating system.

Discussion

Using a novel combination of DNNs trained to perform standard plane recognition and biometry measurement, the AI system was able to automatically perform the entire fetal biometry assessment process using ultrasound videos. This is a novel finding as it integrates DNNs in a live workflow fashion, which contrasts with most of the prior systems, which focus on fetal biometry using datasets of static standard plane images selected by humans, such as Burgos-Artizzu et al. However, it is not the first attempt to implement real-time AI recognition of standard planes in fetal ultrasound videos, which was performed in 2017 by Baumgartner et al. [15]. Our system goes beyond standard plane recognition as it combines automated biometry measurement, measurement fusion, and fetal weight estimation from videos, creating an all-in-one solution. This is a critical distinction because evaluating standard plane recognition from videos, for a complete automated fetal biometry system, requires differentiating between actual standard planes and non-standard planes, which are often in adjacent anatomical positions. In contrast, a still-image dataset only enables a more straightforward classification task—differentiating between predefined standard plane categories (e.g., AC vs. FL vs. HC), which does not fully capture the complexity of routine clinical workflow.

The temporal validation methodology we used is innovative: training and testing datasets are isolated in time, and models are developed using only the training dataset. This is a significant strength of our work, as using the same dataset for both tasks would compromise the data interpretation quality due to overfitting. Additionally, even though the system was trained on still images, it was validated on videos from routine clinical practice. The automatic classification, segmentation,

and weight measurements were performed on the validation videos that contained no pause that could give any hint to the AI as to the correct standard plane images.

Temporal validation has been proven helpful in other medical domains, such as prognosis prediction after a Transcatheter Aortic Valve Implantation (TAVI) or in prediction models of post-partum hemorrhage [16] (Venkatesh et al.).

We used Deep Learning, the best AI technology today, in the fields of image recognition, segmentation, and classification. It is promising as a tool to assist clinicians in performing related repetitive tasks, such as amniotic fluid volume assessment [17] or to assist the less experienced sonographers [18] during the learning process.

The fetal biometry learning curve is relatively long: it is reported that an average of 200 examinations is required to reach technical proficiency, according to fetal sonography quality guidelines [19]. After 2 years of experience, approximately 70 % of clinicians achieve an error margin within 10 % of the birth weight. Consequently, human errors due to insufficient experience can impact the quality of care. To prevent this, often a more experienced clinician is required to double-check or even reperform the examination [20].

The 95 % limits of agreement for the intra-observer difference expressed as a percentage (or z score) can be as high as 3.2 % for HC, 5.4 % for AC and 6.8 % for FL respectively [4]. In that study, the inter-observer differences were 5.1 % (1.00) for HC, 8.9 % (1.37) for AC and 12.0 % (1.60) for FL. As Dudley et al. state, HC measurement is challenging because the key anatomical landmarks are more challenging to identify for less experienced sonographers [21]. According to the INTERGROWTH consensus quality standard, the 95 % confidence interval comprises a variability of 5 to 8 % for HC. This could explain why AI outperformed the beginner sonographer for HC measurements. Surprisingly, medium-level sonographer outperformed AI for FL measurement, while it is expected to be a pretty simple measurement using only two landmarks. One hypothesis is that AI learnt to put calipers on the highest contrast border without considering the quality of the FL image: since the femur is supposed to be inclined horizontally for an optimum measurement. Plus, contrast may be difficult to decipher between the skin and bone if the device parameters are not set on purpose to magnify it.

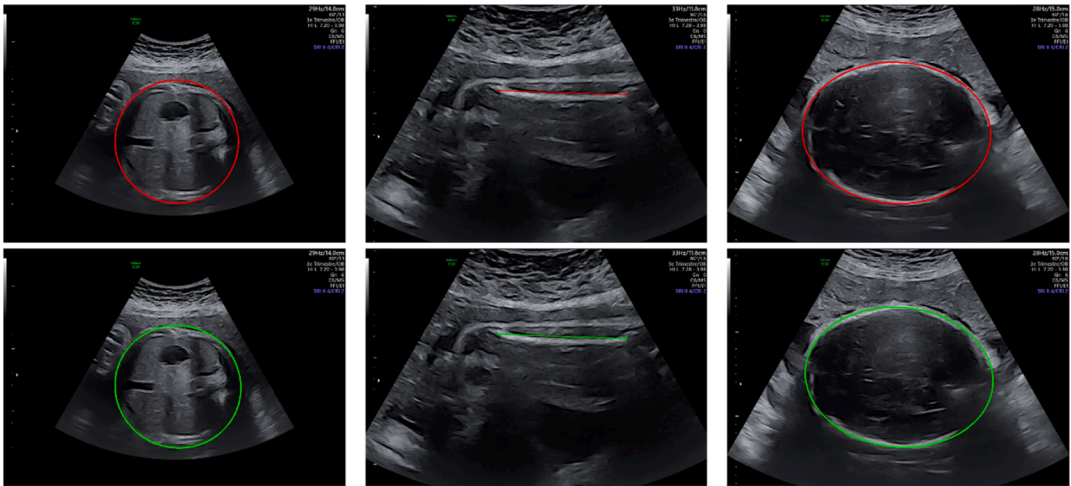
When compared to the expert sonographer, the AI model showed a very higher agreement, with differences as low as 0.96 % (S.D. 0.79 %) for HC, 1.56 % (S.D. 1.39 %) for AC, 1.77 % (S.D. 1.46 %) for FL and 3.10 % (S.D. 2.74 %) for fetal weight estimation. This is promising and would serve the purpose of providing quality routine care to support a range of sonographer skill levels.

Our study has several limitations. Firstly, a single operator acquired the validation videos from a single center and, therefore, did not capture variations across operator scanning techniques or devices. Nevertheless, we observed a variability in the performance of the two controls compared to the Gold Standard measurements obtained by the expert. That may be explained by their different clinical experience, as discussed above.

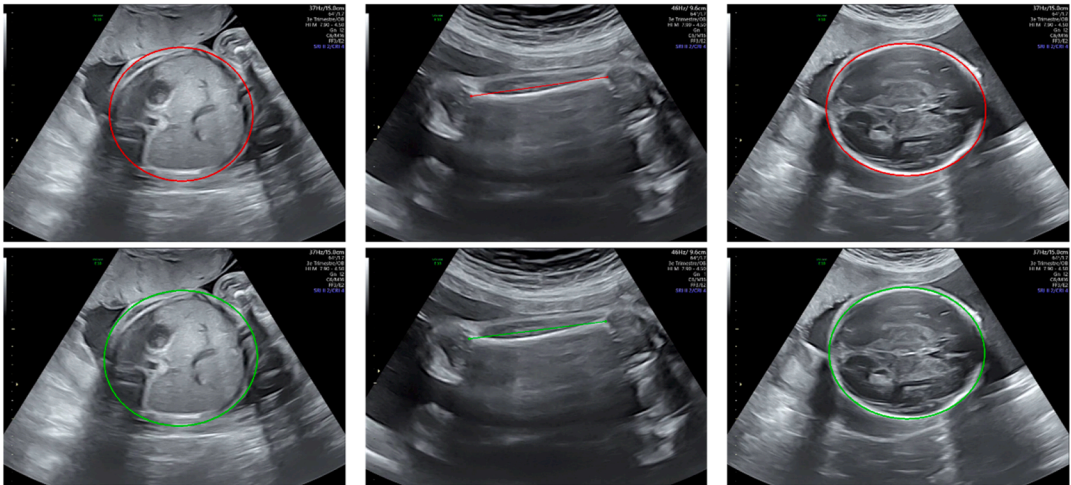
Another limitation is that the expert's measurements were used as the Gold Standard, despite potential variability between experts. A more robust validation could be performed using a cohort of experts. In particular, a gold standard for HC and AC measurements could be obtained using algorithms such as STAPLE [22], which is a well-known technique to combine segmentations from multiple experts.

Another limitation of this study is that we did not investigate the influence of the size of the training set on model performance. A follow-up study is required to characterize the learning rates of the DNNs, and to identify the plateau point where more training data has no significant impact on improving performance.

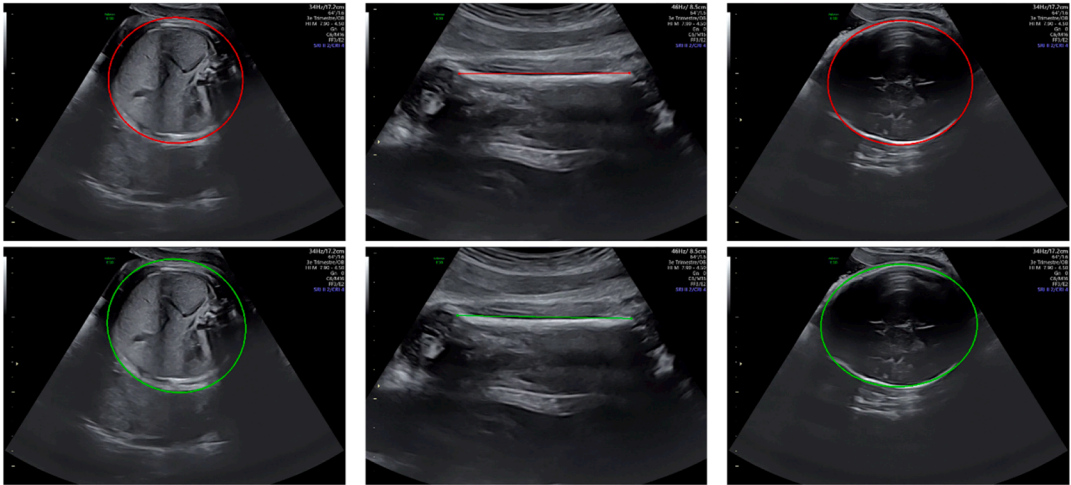
Another limitation is that our study validated the performance of the system in cases when a standard plane was present in the video sequence. In clinical practice, the most challenging part is obtaining the standard plane while scanning the fetus. With this perspective in mind,



Results for patient with highest agreement in fetal weight between AI and expert (0.032% error)



Results for patient with the median agreement in fetal weight between AI and expert (2.10% error)



Results for patient with the lowest agreement in fetal weight between AI and expert (11.50%)

Fig. 5. Standard plane choice and measurements from the AI system and expert obstetrician-gynecologist sonographer, including the results with the patient for which the AI system had the highest agreement in fetal weight estimation with the expert, so as the median and lowest agreement. Annotated images are abdominal 2D ultrasound images: axial plane for AC and HC, frontal plane for FL.

authors like Grandjean et al. have investigated 3D scanning and automatic plane recognition to alleviate bias induced by human operators in manipulating the probe.

Another potential limitation is the use of one weight estimation formula (Hadlock IV formula), which, although used in daily clinical practice, may itself be inherently imprecise due to the reliance on only three biometry parameters. Monier et al. sought to compare the most routinely used formulas: INTERGROWTH formula (IGW) and Hadlock formula. IGW overestimated birthweight at 22–23 weeks compared to Hadlock, but it underestimated birthweight after 28 weeks, which illustrates the difficulty of choosing the appropriate formula according to the clinical context. Espinola-Sánchez et al. [23]. compared different formulas more adapted to machine learning, with the Tabular Prior-data Fitted Network (TabPFN). The primary endpoint was defining the model that produced the estimates closest to the actual weight. However, comparing the performance of a four parameter model TabPFN (R^2 0846) versus Hadlock IV (R^2 0806), it is questionable to state if that difference is clinically relevant.

The AI system could be adapted to and evaluated using other formulae, such as those of Shepard *et al.* and IGW. However, comparing the performance of AI using those different formulas may not solve the underlying problem of the formula.

Another study limitation is that the DNNs were trained on healthy fetuses. The question of whether performance is affected in e.g. fetal growth restriction and macrosomia detection requires investigation. As several authors emphasize, integrating automatic fetal biometry during the ultrasound examination may help reduce human error and variability, allowing more time towards clinical examination, interactions with the patient [24], or anatomy analysis. We believe that our system may help less experienced sonographers evaluate their performance and lessen the rate of medical errors triggered by inaccurate fetal weight estimation.

Moreover, there is today limited access to trained obstetric sonographers in less developed countries [25]. For instance, most women in sub-Saharan Africa undergo pregnancy without a single ultrasound examination [26]. This can impair the quality of medical decisions taken by the obstetrician, for whom knowing gestational age is essential. To address this, in Kenya in 2020. Etyang et al. [27]. tested TraCer, a model to assess gestational age using the head circumference and trans-cerebellar diameter. Sonographers obtained satisfying measurements with the help of this tool, which included a short training video. Using such technology, appropriate care and monitoring may be provided despite the severe shortage of highly trained sonographers.

Conclusion

This work has presented an innovative AI system for automatic fetal biometry using AC, FL, and HC standard plane images, using ultrasound videos and temporal validation methodology. Our model could be helpful in more clinician-deprived contexts and prevent the clinician from being distracted by biometry assessment and focus more on doctor-patient interactions, pathological findings and decision-making. The main future research challenge that remains is how to adapt the AI system to help guide the sonographer to locate each standard plane in real-time.

GLOSSARY

- Machine Learning (ML):

A subfield of artificial intelligence (AI) focused on developing algorithms and models that learn from data to make predictions or decisions. Rather than being explicitly programmed for specific tasks, ML models adapt and refine their performance by identifying patterns and relationships within data, often improving until they reach a level of optimal performance based on the data they are trained on ('training

data').

- Supervised Learning:

An approach to perform ML in which the model is trained on labeled data, meaning that the input data is paired with the correct output. The model learns to map inputs to the correct outputs by minimizing the difference between its predictions and the actual values during training.

- Artificial Neural Network (ANN):

A computational model used in ML that is inspired by the structure and functioning of the human brain. It consists of interconnected nodes, or 'artificial neurons', organized into layers: an input layer that receives data, one or more hidden layers that process and identify patterns in the data, and an output layer that produces the final network output.

- Deep Neural Network (DNN):

A type of ANN with two or more hidden layers between the input and output layers. A DNN's depth refers to the number of hidden layers in the model and increasing the depth typically allows the model to learn more complex, hierarchical representations of the data. This property makes DNNs particularly effective for tasks such as image recognition, speech processing, and natural language understanding.

- Deep Learning (DL):

A subset of ML that specifically uses DNNs, often requiring large datasets to train such models adequately.

Declaration of competing interest

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jogoh.2025.103039](https://doi.org/10.1016/j.jogoh.2025.103039).

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